



Institute of Engineering and Technology (IET)

Major Project Synopsis (PR1107)

HealthTwin

Digital Twin for Remote Patient Monitoring

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FACULTY GUIDE

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1. Introduction

Continuous monitoring of high-risk and chronic patients is a persistent healthcare challenge clinical staff cannot manually track every patient in real time, and conventional threshold-based alarms generate a large share of false or clinically irrelevant alerts, causing alarm fatigue. HealthTwin proposes a digital twin system for predictive remote patient monitoring: a continuously updated virtual representation of a patient's physiological state, paired with a Transformer-based time-series model that forecasts deterioration risk hours in advance, an agentic decision-support layer that raises alerts only when genuinely warranted, and a secure communication channel so the clinician can directly reach the patient. The project uses public, de-identified ICU datasets (MIMIC-III, PhysioNet) for development and evaluation.

2. Objectives

- Design a digital twin representation of a patient's physiological state from vital-sign data.
- Train a Transformer-based model to predict deterioration risk hours in advance.
- Build an agentic layer that raises alerts only when genuinely warranted, reducing false alarms.
- Integrate prediction, decision-support, communication, and a visualization dashboard into one pipeline.
- Evaluate performance using AUROC, sensitivity/specificity, and early-warning lead time.

3. Key Features

- **Digital Twin Representation** - Continuously updated virtual model of the patient's vitals and trends.
- **Transformer-Based Risk Prediction** - Self-attention model forecasting deterioration hours ahead.
- **Intelligent Alerting Agent** - Confidence-aware agent that filters predictions into no-action / soft-notify / urgent alert, cutting false alarms.
- **Monitoring Dashboard** - Real-time (simulated) view of twin state, risk trend, and alert history.
- **Doctor–Patient Communication** - Secure messaging plus an embedded video-call link (Jitsi Meet), triggered directly from the alert workflow.

4. Technology Stack

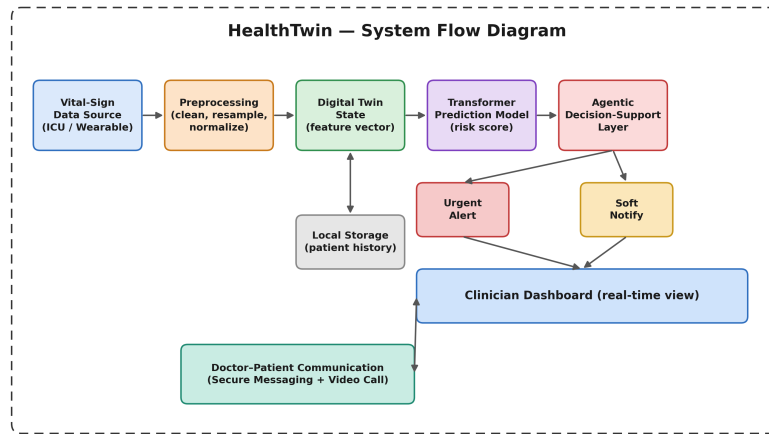
- **Data Processing** - Python, Pandas, NumPy.
- **Predictive Model** - PyTorch + HuggingFace Transformers (Temporal Fusion Transformer).
- **Decision Layer** - Lightweight Python rule-guided agent.
- **Storage** - SQLite.
- **Communication** - In-app messaging (SQLite-backed) + Jitsi Meet embed for video calls.
- **Visualization** - Streamlit dashboard.
- **Datasets** - MIMIC-III, PhysioNet/CinC 2019 Sepsis Challenge.

5. Methodology

1. Data Acquisition & Preprocessing - clean, window, and normalize public ICU vital-sign data.
2. Digital Twin State Design - represent each patient as a continuously updated feature vector.
3. Model Development - train the Transformer-based predictor on windowed vitals; validate on held-out data.
4. Decision-Support Logic - build the agent that turns predictions into alert actions.
5. Communication Module - implement secure messaging and an embedded video-call link, wired into the alert workflow.
6. Integration - connect all stages into one Docker-packaged application.
7. Evaluation - benchmark against a threshold-based baseline using AUROC, sensitivity, and lead time.

6. System Flow Diagram

The diagram below summarizes the end-to-end flow of data through the HealthTwin pipeline, from raw vital-sign input to the clinician-facing alert.



7. Timeline

- **August (Week 3–4)** - Literature review, dataset selection, requirement finalization.
- **September (Week 1–2)** - Data preprocessing pipeline and exploratory analysis.
- **September (Week 3–4)** - Digital twin representation and baseline model.
- **October** - Transformer model development and tuning.
- **November (Week 1–2)** - Agentic decision layer and evaluation.
- **November (Week 3–4)** - Dashboard development and integration.
- **December (Week 1–2)** - Testing, documentation, and report writing.
- **December (Week 3–4)** - Final demo and presentation.

8. Expected Outcomes

- A working HealthTwin prototype (prediction + alerting + communication + dashboard).
- A trained Transformer model with measurable early-warning performance.
- Demonstrated reduction in false/non-actionable alerts vs. a threshold baseline.
- A reproducible, end-to-end pipeline with an integrated doctor–patient communication channel.

9. Future Enhancements

- Real wearable/IoT sensor integration.
- Cloud deployment for multi-patient, hospital-scale use.
- Integration with electronic health record (EHR) systems.
- A companion mobile app for clinician alerts.

10. Conclusion

HealthTwin demonstrates how a digital twin, combined with a Transformer-based predictive model and an intelligent alerting layer, can shift patient monitoring from reactive to proactive addressing alarm fatigue and supporting earlier clinical intervention. Built as a self-contained prototype, it lays a clear foundation for future extension into real hardware and larger-scale deployment.

11. References

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